Eric J. Sung — Curriculum Vitae

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Contact Information

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Location: Tucson, AZ
Citizenship: USA

Education

University of Arizona, Tucson, AZ — Expected May 2028
Doctor of Philosophy, Applied Mathematics
Advisor: Dr. Charles A. Stafford

Tulane University, New Orleans, LA — May 2021
Master of Science, Applied Mathematics
Master's Thesis: *Mathematics and Applications of Feynman Diagrams*
Advisor: Dr. Victor H. Moll

Tulane University, New Orleans, LA — May 2020
Bachelor of Science, Mathematics
Minor: Geology

Publications

[1] **E. J. Sung**, A. G. Campos, H. Abele, and D. I. Bondar, "Decoherence-free entropic gravity for a Dirac fermion", Physical Review D **108**, 104036 (2023).

Under Review

[1] **E. J. Sung** and C. A. Stafford, *Decoherence and the Reemergence of Coherence From a Superconducting "Horizon"*, Mar. 2026, arXiv:2603.16765 [quant-ph].

[2] **E. J. Sung**, B. Koch, T. Jenke, H. Abele, and D. I. Bondar, *Generalized Boundary Conditions for the qBounce Experiment*, Oct. 2025, arXiv:2510.15341 [quant-ph].

In Preparation

[1] "From clutter to context: a taxonomy for automatic target recognition", Manuscript in preparation.

Presentations

[1] C. A. Stafford and **E. J. Sung**, "Decoherence from a superconducting "horizon"", in APS Meetings Archive, APS Global Physics Summit 2026, Abstract W32.00011 (American Physical Society, Mar. 2026).

[2] **E. J. Sung**, "Decoherence and the Reemergence of Coherence From a Superconducting "Horizon"", in Program in Applied Mathematics Brown Bag Seminar, The University of Arizona, Tucson, AZ (Mar. 2026).

[3] **J. Sung**, B. Crooke, D. Carstens, C. Woock, R. Amer, and A. Kolker, "Sentinel-1A InSAR Time Series Analysis for Monitoring of Surface Subsidence in Coastal Louisiana", in AGU Fall Meeting Abstracts, Vol. 2018 (Dec. 2018), G21C–0571.

Conferences and Workshops

- 2026 MSTIC Summer Symposium (Talk) — Aug 2026
  Sandia National Laboratories, Albuquerque, NM
- Quantum Thermodynamics & Decoherence (QTD) Symposium 2026 (Poster) — Jun 2026
  University of Rochester, Rochester, NY
- American Physical Society Global Physics Summit 2026 (Talk) — Mar 2026
  Denver, CO
- Math for All in New Orleans — Mar 2020
  Tulane University, New Orleans, LA
- Clifford Lectures — Oct 2019
  Tulane University, New Orleans, LA
- 2018 American Geophysical Union (AGU) Fall Meeting (Poster) — Dec 2018
  Washington, DC

Awards and Honors

- Best Graduate Summer Student Presentation, Sandia National Laboratories — Aug 2026
- QTD Symposium Entropy Whisperer Poster Award (100 USD) — Jun 2026
- QTD Symposium Travel Award (1000 USD) — Jun 2026
- Herbert E. Carter Travel Award (600 USD), Graduate College UArizona — Jan 2026
- Sigma Xi, The Scientific Research Honor Society, Full Member — Dec 2025–Present
- University of Arizona Graduate College Fellowship — Aug 2023

Research Experience

Sandia National Laboratories, Albuquerque, NM

Year-Round Graduate Intern — August 2026–Present

R&D Graduate Summer Intern — May 2026–August 2026

University of Arizona, Department of Physics, Tucson, AZ

Graduate Research Assistant — Dec 2024–Present

- Developing a superconducting analog of black hole physics using non-equilibrium Green's functions (NEGFs) with Prof. Charles A. Stafford
- Calculating NEGF transmission spectra for superconductor–normal devices (Landauer-Büttiker conductance) via simulations on Python and developing dephasing–probe extensions to estimate electronic phase-coherence time
- Implementing grid search algorithms on UA's high-performance computing cluster to optimize superconducting parameters to increase transmission and phase coherence time
- Discovered solid-state analog of horizon-induced decoherence and novel reemergence of coherence
- First-author paper preprint submitted

University of Arizona, Department of Mathematics, Tucson, AZ

Graduate Research Assistant — May 2024–May 2025

- Implemented reinforcement learning algorithms using PyTorch to design optimal control strategies for particle accelerators under Prof. Michael Chertkov
- Developed a differentiable neural ODE surrogate of the accelerator beamline using training data generated with LANL's HPSim, enabling rapid emulator-based testing of control ideas
- Designed and optimized particle accelerator simulations on UA's high-performance computing cluster using CUDA for efficient parallel computations

Independent Studies Researcher — Sep 2023–Dec 2024

- Investigated quantum trajectories and decoherence of Schrödinger cat states under Prof. Janek Wehr
- Designed quantum Monte Carlo simulations of Schrödinger cat states in optical cavities using QuTiP on Python (see GitHub repo)
- Worked on formulating Weierstrass-Enneper representations of the Dirac equation with Prof. Patrick Shipman, discovered a general symmetry theorem

Army Educational Outreach Program Undergraduate Internship, **Tulane University, Department of Physics, New Orleans, LA**

Research Intern — May 2023–Aug 2023

- Collaborated with Prof. Denys I. Bondar and the Institut Laue-Langevin to investigate standard gravity discrepancies of free-falling neutrons observed in the 2023 qBounce experiment
- Discovered, via open quantum systems simulations using QuTiP on Python and Mathematica, that the experimental boundary conditions were responsible for the discrepancy
- First-author paper preprint submitted

Tulane University, Department of Physics, New Orleans, LA

Research Assistant — May 2021–Aug 2023

- Conducted research with Prof. Denys I. Bondar on entropic gravitational effects on Dirac fermions in an open quantum systems framework
- Formulated a Lindblad master equation for free-falling Dirac fermions, demonstrating coherence under strong gravitational coupling and aligning with experimental results
- Discovered a novel zitterbewegung gravitational effect via open quantum systems simulations using Python and Mathematica; published findings in first-author paper in Physical Review D

Tulane University, Department of Mathematics, New Orleans, LA

Master's Research — Aug 2020–May 2021

- Conducted master's thesis research, under Prof. Victor H. Moll, on the Feynman diagrams and quantum field theory (QFT) along with their applications

- Developed proficiency in the application of QFT, scattering theory, electrodynamics, and Lie theory
- Successfully defended thesis to thesis committee and submitted written master's thesis to Tulane University Thesis & Dissertations Archive

Tulane University, Department of Physics, New Orleans, LA
Research Assistant May 2020–Aug 2020

- Investigated the accuracy and applications of the Equations of Motion (EOM) method under Prof. George Rosensteel
- Designed and implemented EOM method using Python to the Morse potential, calculating excitation energies in higher dimensions with high precision
- Gained proficiency in Python and computational physics

Tulane University, Department of Mathematics, New Orleans, LA
Senior Seminar Research Aug 2019–May 2020

- Conducted year-long research under Prof. Lisa J. Fauci investigating the collective phase of *Bacillus subtilis* bacteria in viscous flows
- Implemented regularization algorithms on Python and MATLAB to compute forces and velocities of bacterial motion to determine the origin of collective phase
- Completed an expository paper, *Mathematical Fluid Dynamics of Microorganisms*, and gave a presentation of results to Senior Seminar class

Tulane University, Department of Earth & Environmental Sciences, New Orleans, LA
Research Assistant Jun 2018–Dec 2018

- Investigated land subsidence in Golden Meadows, Coastal Louisiana, under Prof. Reda Amer
- Collected and analyzed over 100 Real-Time Kinematic GPS elevation points and Sentinel-1A satellite InSAR remote sensing data
- Implemented interpolation and image classification algorithms on ArcGIS and ENVI, discovering a 2.91% land reduction (90 km^2) from 1985-2017 at 0.2-1.64 cm/year
- Presented findings as a first-author poster at the Fall 2018 American Geophysical Union in Washington D.C.

Teaching

University of Arizona, Department of Mathematics, Tucson, AZ
Graduate Teaching Assistant Aug 2023–Present

- Solo instructor for Math 112 College Algebra
- Graduate teaching assistant for Math 186J College Algebra Supplemental Instruction Seminar, Math 196V Vector Calculus Supplemental Instruction Seminar, and Math 468 Applied Stochastic Processes
- Hosted weekly office hours, graded homework assignments, and proctored exams

Projects

Predicting Grant Success: A Multimethod Approach Using TDA, NLP, and Bibliometric Analysis
Paper, GitHub Repo Oct 2024–Dec 2024

- Developed and implemented machine learning models (Logistic Regression, Random Forest, and Gradient Boosting) using R and Python's scikit-learn to predict Department of Defense grant approval rates based on bibliometric data
- Enhanced prediction accuracy by fine-tuning hyperparameters for Random Forest and Gradient Boosting models, achieving a Test AUC-ROC of 0.743 and improving overall classification accuracy to 71.3%
- Analyzed feature importance and identified i10-index and total citations as the most influential predictors, providing actionable insights into grant success factors for academic researchers

Numerical Solutions to Reaction Diffusion Equations
Paper Oct 2020–Dec 2020

- Investigated long-time dynamics and pattern formation in 1D/2D reaction–diffusion PDEs under varied boundary conditions

- Built MATLAB finite-difference solvers using Crank–Nicolson (diffusion), ADI (2D), and explicit multistep updates for nonlinear reaction terms
- Implemented Dirichlet, Neumann (ghost-node), and periodic BCs with efficient tridiagonal/LU linear solves to ensure stable, scalable time-stepping
- Reproduced Gray–Scott two-species pattern regimes; generated heatmaps/cross-sections and documented parameter sensitivity and emergent structures in a formal report

Walter L. Cohen High School, New Orleans, LA
Tutor Aug 2016–Dec 2016

- Taught math, general sciences, English, and history at Walter L. Cohen High School
- Collaborated with other teachers in designing and implementing new teaching methods and problem sets

Leadership and Outreach

University of Arizona, Tucson, AZ
Graduate Student Peer Mentor August 2026–Present

- Served as a peer mentor for first-year graduate students in the applied mathematics PhD program, providing guidance on academic and research opportunities
- Hosted monthly meetings to discuss progress with mentees

Tulane University Earth Science Club, New Orleans, LA
President May 2018–Dec 2018

- Established and led a student earth science club, growing membership to 40 students
- Secured up to $1000 in funding through successful grant proposals to the undergraduate student government for supporting club activities
- Organized and coordinated events, including hiking trips, camping excursions, and networking workshops

Skills

Programming/Software: Python (pandas, NumPy, SciPy, SymPy, Scikit-learn, TensorFlow, PyTorch, QuTiP), Julia, R, Mathematica, CUDA, MATLAB, C++, SQL, HTML, CSS, JavaScript, Git, Microsoft Office, ArcGIS, ENVI, Jupyter, LaTeX
Languages: Korean (Elementary), Spanish (Elementary)

Professional Societies

American Physical Society (APS) Aug 2024–Present
Society for Industrial and Applied Mathematics (SIAM) Jan 2022–Present
American Mathematical Society (AMS) Jan 2024–Present

References

References available upon request